

Deveshwar Hariharan

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EDUCATION

Ph.D. and M.S., Electrical Engineering

North Carolina State University

May 2026

Raleigh, NC, USA

- GPA: 3.89/4.0

Bachelor of Technology, Electrical Engineering

Amrita Vishwa Vidyapeetham, Ettimadai Campus

May 2018

Coimbatore, Tamil Nadu, IN

- GPA: 3.8/4.0

SKILLS

Expertise: Computer Vision, Machine Learning, Deep Learning, Cooperative Perception, Systems Engineering

Languages: Python, C++, MATLAB, CUDA

Libraries and Tools: ROS, TensorFlow, OpenCV, PyTorch, PCL, TensorRT, CARLA, Gazebo, KiCad

Management and Leadership: IRB Compliance, Cross-Functional Leadership, Advocacy, Governance

EXPERIENCE

Lead Graduate Research Assistant

EcoPRT Autonomous Vehicle Lab, North Carolina State University

June 2019 – present

Raleigh, NC

- Lead electrical and computer science teams in autonomous vehicle development lab
- Develop and deploy object recognition algorithms to improve prediction accuracy and blindspot maneuvering using 3D point-clouds and cooperative perception
- Manage vehicle code to industry standards to improve runtime speed, code cross-compatibility, and to streamline research-to-test processes
- Design custom deployment PCBs to save the lab thousands of dollars in deployment costs for improved IMUs
- Lead 15 software engineering and robotics undergraduate students in developing a Fleet Management Server
- **Skills:** Computer Vision, Machine Learning, Cooperative Perception, ROS, C++, PCB Development, API Development, Cross-Functional Leadership

Student Researcher

EcoPRT Autonomous Vehicle Lab, North Carolina State University

Jan 2019 – June 2019

Raleigh, NC

- Developed the software for emergency braking upon detection of obstacles for autonomous vehicle
- Implemented a computer vision model and tracking algorithm to recognize obstacles
- Developed safety tests to pass IRB requirements to test autonomous vehicles on campus
- **Skills:** Computer Vision, Machine Learning, ROS, C++, Point Cloud Library (PCL), IRB Compliance

COMMUNITY & LEADERSHIP

President of Graduate Student Body

Graduate Student Association (GSA), NC State University

May 2021 – April 2023

Raleigh, NC

- Created and funded Graduate Student Parent Association (GSPA)
- Collaborated to advocate for childcare and maternity/paternity benefits for graduate students
- Introduced and passed an amendment to the GSA Constitution to increase funding from GSA to Special Interest Groups (SIGs)
- Worked with SIGs to take advantage of the new amendment and increase funding
- Collaborated with peers from MIT and CMU in garnering support for the creation of the GRAD Caucus in the US House of Representatives
- Enlisted NC State University's GSA in the advisory team to the GRAD Caucus

PROJECTS

- Home Automation Projects** | *Docker, Python, RaspberryPi, PlatformIO, ESPHome, Ollama* **Present**
- Build custom mmWave radars for presence and occupancy detection
 - Deploy OSS tools for automation, logging, and control of IoT devices using Home Assistant
 - Deploy fully local LLM-based voice assistants in multiple natural languages
 - Build and integrate custom HiFi speakers
 - Implement custom audio-sync protocol (Sendspin) to synchronize audio across several speakers
- Automatic Cat Feeder** | *Python, HTML, CSS* **June 2020**
- Solved the problem of being woken up by the cat early in the morning
 - Built a custom cat feeder with automated feeding times, voice commands, and perception algorithm
 - Now the cat goes to the feeder when he is hungry and it's feeding time, and we get to sleep
- Line Following Robot** | *Python, OpenCV* **Aug 2019 – Dec 2019**
- Developed a control system from scratch to maximize the time speed of the robot without deviating from the track
- Lane Following drone** | *ROS, Gazebo, MavROS, Deep Learning* **Jan 2019 – May 2019**
- Built and wired a drone from scratch
 - Developed the ROS and Gazebo interfaces with MavROS for the flight controllers
 - Developed and implemented a ROS node with a convolution neural network-based lane detector
 - Developed the control systems for the vehicle to change course based on the lane predictions

AWARDS

- Graduate Student of the Year, Leadership** **2023**
Electrical and Computer Engineering Department, NC State University *Raleigh, NC*
- Student of the Year, Academic** **2015**
Electrical and Electronics Engineering, Amrita Vishwa Vidhyapeetham *Coimbatore, Tamil Nadu, India*

PUBLICATIONS

- **Hariharan, D.**, Zhu, Y., Pulithaya, D. U., Yoon, M., Hollar, S. (accepted) "RinneFormer - Cross Attention in Cooperative Perception", In review *Intelligent Vehicles Symposium, 2026*
- **Hariharan, D.**, Patil, N., Issa, J., Hollar, S., "Improvements to a Novel Autonomous Personal Rapid Transit (PRT) Prototype Vehicle", In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, 2024*
- Chu, Y., **Hariharan, D.**, Hollar, S., Feng, F., "What Does That Car Mean? The Influence of Vehicle Motion and Symbolic Patterns of LED Signals on Pedestrians' Interpretation of a Vehicle's Intent", *Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 2022*
- **Hariharan, D.**, PointPillars++, A Spatial Feature Encoder for Classification of Raw Point Cloud Data from LiDARs, *Electronic Thesis, NC State University 2022*
- **Hariharan, D.**, Venkat, G., Vasudevan, S. K., "A Novel Approach for increased transaction security with Biometrics and One Time Password A complete implementation.", *International Journal for Advanced Intelligence Paradigms, 2024*